

CHAPAIS, QC, Canada

WMO: 718240

Lat: 49.8225N

Lon: 74.9753W

Elev: 1250

StdP: 14.04

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-25.8	-39.1	0.6	-31.0	-33.0	0.9	-25.7	12.2	10.3	10.9	7.5	2.4	210	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	82.5	64.1	78.9	62.1	75.6	60.9	68.2	75.8	66.4	73.3	64.4	71.2	6.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	100.3	70.3	63.9	93.2	69.0	61.9	86.7	67.4	33.3	76.3	31.7	73.1	30.2	71.0	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.7	8.7	-39.9	88.8	4.4	2.8	-43.1	90.8	-45.6	92.4	-48.1	93.9	-51.2	96.0		
			-40.4	72.4	4.7	2.3	-43.8	74.0	-46.6	75.3	-49.2	76.6	-52.7	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.1	-0.3	2.6	14.6	30.5	46.8	57.3	61.6	59.3	51.5	38.8	24.6	8.6
	DBStd	24.25	14.90	13.66	14.03	10.56	9.75	7.87	6.12	6.29	8.12	8.61	10.13	13.27
	HDD50	7269	1561	1326	1097	588	183	21	1	6	77	365	762	1282
	HDD65	11717	2026	1746	1562	1036	570	254	143	198	410	813	1212	1747
	CDD50	1117	0	0	1	2	82	239	360	293	123	17	0	0
	CDD65	89	0	0	0	0	4	22	37	20	6	0	0	0
	CDH74	1053	0	0	0	0	99	333	407	159	55	0	0	0
	CDH80	226	0	0	0	0	23	93	86	18	6	0	0	0
	WSAvg	4.1	3.9	4.1	4.4	4.5	4.3	3.9	3.7	3.6	3.9	4.1	4.3	3.8
	PrecAvg	38.9	2.2	1.7	1.9	2.2	2.9	3.8	4.6	4.2	4.8	3.8	3.6	2.7
Precipitation	PrecMax	47.8	3.9	3.5	3.9	3.7	5.4	6.5	7.4	9.1	11.2	6.7	6.7	5.3
	PrecMin	31.7	0.5	0.0	0.0	0.5	0.7	1.3	1.7	1.6	2.4	1.5	1.4	0.0
	PrecStd	4.5	0.9	0.8	0.9	0.8	1.2	1.5	1.6	1.7	1.9	1.4	1.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	36.0	52.0	63.1	82.7	87.8	86.8	82.6	79.8	68.5	50.9	38.5
		MCWB	36.6	33.5	43.0	48.0	62.0	65.1	66.0	67.4	63.9	57.4	47.5	37.5
	2%	DB	32.9	32.1	44.5	56.9	75.9	82.3	82.2	78.3	74.4	62.0	45.6	33.5
		MCWB	32.5	31.1	36.7	43.5	55.8	62.3	64.5	64.0	62.7	54.1	43.2	33.0
	5%	DB	27.5	28.3	39.1	50.9	70.4	77.4	78.7	74.9	69.8	56.6	41.1	31.1
		MCWB	26.7	27.0	33.3	40.6	53.8	60.2	62.5	62.3	60.7	52.0	39.5	30.5
	10%	DB	22.5	24.5	34.8	45.8	64.9	73.0	74.9	71.5	65.7	52.1	37.2	27.6
		MCWB	21.7	23.1	31.2	38.0	51.0	58.3	61.7	61.4	58.7	48.5	35.9	27.0
	0.4%	WB	36.6	33.6	45.8	51.0	65.5	69.8	71.1	69.9	67.7	59.7	49.1	38.0
		MCDB	37.0	35.4	47.9	57.8	74.0	78.8	78.1	77.1	75.4	66.6	50.3	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.7	30.6	38.2	45.5	59.9	66.5	68.2	67.5	65.2	55.5	44.0	33.2
		MCDB	33.2	31.7	42.6	52.7	69.2	75.3	75.7	73.7	70.9	59.5	45.1	33.6
	5%	WB	26.3	26.8	34.8	42.2	56.1	63.7	66.1	65.7	62.5	51.9	39.5	30.7
		MCDB	27.1	28.0	38.4	49.5	65.8	72.3	73.5	71.3	68.3	55.1	41.0	31.2
	10%	WB	21.3	22.8	31.6	39.1	53.2	61.4	64.0	63.4	59.5	48.4	35.9	27.1
		MCDB	22.1	24.0	35.1	45.7	63.0	70.0	71.1	69.0	65.2	50.9	37.2	27.6
Mean Daily Temperature Range	5% DB	MDBR	19.7	21.8	23.9	20.6	23.5	23.8	21.8	19.6	19.3	13.3	11.9	16.2
		MCDBR	18.1	16.6	26.0	27.7	31.6	31.1	28.3	25.4	24.8	21.7	14.7	13.2
		MCWBR	17.9	15.0	19.3	16.4	15.5	14.2	14.1	13.6	15.1	14.6	13.1	13.5
		MCDBR	18.3	15.6	21.9	24.1	27.0	24.0	21.5	20.0	21.7	17.9	14.4	14.0
	5% WB	MCWBR	18.3	14.7	17.8	16.1	15.8	12.9	13.2	13.1	14.4	13.4	13.4	14.0
Clear-Sky Solar Irradiance	taub		0.247	0.263	0.279	0.329	0.337	0.356	0.373	0.354	0.348	0.304	0.280	0.243
	taud		2.285	2.257	2.263	2.339	2.476	2.452	2.424	2.481	2.466	2.581	2.487	2.340
	Ebn at Noon		258	282	296	291	292	286	279	280	270	267	247	244
	Edh at Noon		22	29	34	36	33	34	35	31	29	21	18	18
All-Sky Solar Radiation	RadAvg		388	727	1204	1528	1661	1784	1620	1327	1004	565	283	232
	RadStd		49	92	112	119	155	137	155	91	96	66	35	36

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	+3.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page